Deep Learning Curriculum

8-Week Beginner-to-Intermediate Learning Roadmap

Duration	8 Weeks
Recommended Study Time	10–15 hours per week
Level	Beginner to Intermediate
Primary Language	Python
Core Frameworks	PyTorch or TensorFlow/Keras
Outcome	Build, train, evaluate, and improve deep learning models

Learning Objectives

- Understand the mathematical and machine learning concepts behind deep learning.
- Build neural networks and understand forward propagation and backpropagation.
- Train models using loss functions, optimizers, validation, and evaluation metrics.
- Apply CNNs to computer vision and RNN/LSTM concepts to sequential data.
- Understand attention and transformer fundamentals.
- Complete portfolio-ready deep learning projects.

Week 1: Foundations and Prerequisites

Weekly Goal: Build the programming, mathematics, and machine learning foundation required for deep learning.

Topics to Learn

• Python review: functions, classes, modules, virtual environments

• NumPy arrays, broadcasting, matrix operations, and vectorization

• Pandas basics and data preprocessing

• Matplotlib for visualization

• Linear algebra: vectors, matrices, dot products, matrix multiplication

• Calculus intuition: derivatives, gradients, and the chain rule

• Probability and statistics: distributions, mean, variance, and normalization

• Machine learning workflow: train, validation, and test sets

Practice: Implement matrix operations with NumPy and build a small data preprocessing notebook.

Weekly Deliverable: Python + NumPy foundations notebook

Week 2: Neural Networks Fundamentals

Weekly Goal: Understand how artificial neural networks learn from data.

Topics to Learn

• Artificial neurons and the perceptron

• Weights, biases, and linear combinations

• Activation functions: ReLU, Sigmoid, Tanh, Softmax

• Single-layer and multilayer neural networks

• Forward propagation

• Loss functions: MSE and Cross-Entropy

• Gradient descent and learning rate

• Introduction to backpropagation

Practice: Create a simple neural network conceptually and implement a basic classifier.

Weekly Deliverable: Neural network fundamentals notebook

Week 3: Training Deep Neural Networks

Weekly Goal: Learn how to train, validate, debug, and improve neural networks.

Topics to Learn

• Backpropagation in practice

• Batch, mini-batch, and stochastic gradient descent

• Optimizers: SGD, Momentum, RMSprop, and Adam

• Epochs, batch size, and learning rate

• Training and validation loss

• Overfitting and underfitting

• Regularization: L1 and L2

• Dropout and Batch Normalization

• Early stopping and learning-rate scheduling

Practice: Train a dense neural network on a classification dataset and compare optimizers.

Weekly Deliverable: ANN classification project with training curves

Week 4: Deep Learning Framework and Project Workflow

Weekly Goal: Use a professional deep learning framework to build complete models.

Topics to Learn

• Choose one primary framework: PyTorch or TensorFlow/Keras

• Tensors and computational operations

• Dataset loading and preprocessing pipelines

• Defining models and layers

• Training loops or high-level training APIs

• Evaluation and prediction

• Saving and loading trained models

• Using GPU acceleration when available

• Experiment organization and reproducibility

Practice: Build an end-to-end image or tabular classification model using your chosen framework.

Weekly Deliverable: Framework-based deep learning project

Week 5: Convolutional Neural Networks (CNNs)

Weekly Goal: Learn deep learning techniques for image and computer vision tasks.

Topics to Learn

• Image representation and channels

• Convolution operations and filters

• Feature maps, kernels, stride, and padding

• Pooling layers

• CNN architecture design

• Flattening and fully connected layers

• Data augmentation

• Image classification evaluation

• Transfer learning introduction

Practice: Train a CNN on MNIST, Fashion-MNIST, or CIFAR-10.

Weekly Deliverable: Image classification CNN project

Week 6: Sequence Models and NLP Basics

Weekly Goal: Understand how neural networks process sequences and text.

Topics to Learn

• Sequential data and time dependencies

• Introduction to RNNs

• Vanishing and exploding gradients

• LSTM architecture and gates

• GRU overview

• Text preprocessing and tokenization

• Word embeddings

• Sequence classification

• Introduction to sentiment analysis

Practice: Build a sentiment analysis or sequence classification model.

Weekly Deliverable: RNN/LSTM-based NLP project

Week 7: Attention and Transformers

Weekly Goal: Learn the core concepts behind modern transformer-based AI systems.

Topics to Learn

• Why attention was introduced

• Attention intuition

• Query, Key, and Value concepts

• Self-attention

• Multi-head attention

• Positional encoding

• Transformer encoder and decoder overview

• Pretrained transformer models

• Fine-tuning versus feature extraction

• Introduction to Hugging Face ecosystem

Practice: Use a pretrained transformer for text classification or another NLP task.

Weekly Deliverable: Transformer-based mini project

Week 8: Transfer Learning, Deployment, and Capstone Project

Weekly Goal: Combine your knowledge into a complete portfolio-ready deep learning application.

Topics to Learn

- Transfer learning and fine-tuning
- Model evaluation and error analysis
- Confusion matrix, precision, recall, and F1-score
- Hyperparameter tuning fundamentals
- Model serialization and inference
- Building a simple interface with Streamlit
- Project documentation and README writing
- GitHub project organization
- Responsible AI basics and model limitations

Practice: Complete one capstone project from problem definition through deployment.

Weekly Deliverable: Complete GitHub-ready capstone project

Recommended Capstone Project Ideas

Project	Skills Applied
Image Classification App	CNNs, transfer learning, data augmentation, Streamlit
Sentiment Analysis System	Text preprocessing, embeddings, LSTM or transformers
Plant Disease Detection	CNNs, transfer learning, computer vision
Handwritten Digit Recognition	Neural networks, CNNs, evaluation
Customer Review Classifier	NLP, sequence models, transformers
Time-Series Forecasting	Sequential data, LSTM/GRU concepts, evaluation

Suggested Weekly Study Pattern

- Day 1–2: Learn theory and take structured notes.
- Day 3: Implement concepts in Python.
- Day 4: Complete exercises and experiments.
- Day 5: Work on the weekly mini-project.
- Day 6: Review errors, improve the model, and document findings.
- Day 7: Revision, rest, or catch-up.

Recommended Technology Stack

- Python
- NumPy and Pandas
- Matplotlib
- Scikit-learn
- PyTorch or TensorFlow/Keras
- Jupyter Notebook
- Hugging Face Transformers
- Streamlit
- Git and GitHub

Expected Outcome After 8 Weeks

After completing this curriculum, you should be able to explain the major concepts of deep learning, prepare datasets, build and train neural networks, use CNNs for images, understand RNN/LSTM models for sequences, apply transformer fundamentals, use transfer learning, evaluate model performance, and complete a documented deep learning project suitable for your portfolio.